
// BIOKEY SYSTEM v2.0

Your heartbeat is the key — at any stress level.

Stress-Invariant Biometric Authentication · AES-256-GCM

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Jigmond Alexandru · Milotin Rareş · Sramec Iosif · Stan Cristian // UPT AC

Authentication breaks when it matters most.

STRESS

A soldier's ECG at 70 bpm is not the same at 160 bpm. The system denies its own operator.

CAPTURE

If the device is seized, all onboard data is immediately accessible. No cryptographic barrier.

COERCION

Passwords and cards can be taken by force. The operator cannot protect them.

SINGLE TEMPLATE

Every existing biometric system enrolls once, at rest. Field conditions break the assumption.

Operator-bound encryption. Stress-invariant.

MULTI-TEMPLATE ENROLLMENT

The operator is enrolled across 5 physiological states – rest to heavy exertion. The system recognizes them at any heart rate, automatically.

BIOMETRIC KEY DERIVATION

The AES-256 encryption key is derived directly from the cardiac signature + PIN. Without the right person, the data is cryptographically inaccessible – not just access-restricted.

SECOND FACTOR – PIN

Even with correct biometrics, a wrong PIN fails at the cryptographic level. Captured biometric data alone is not enough.

Any role where the right person must be at the controls – right now.

DEFENCE

- Drone operator authentication
- Forward operating base access
- Classified device protection
- Two-man rule implementation

CIVIL & INDUSTRIAL

- Critical infrastructure access
- Air traffic controller verification
- Nuclear plant operator auth
- Surgeon / operating room control

Cryptographic layer is production-ready. Path to deployment: TPM/HSM integration – no fundamental blockers.